

Proof of Proposition 2.2

Setting and notation. Let $t = \langle n_1, \dots, n_k \rangle \in \mathcal{T}_M$ and $t' = \langle n'_1, \dots, n'_\ell \rangle \in \mathcal{T}_S$ be traces. Let N be the action and branch elements of M , let N' be the atomic-action and branch spans of T , and let $\mu \subseteq N \times N'$ be a node-match relation. Traces are non-empty, so $|t|$ and $|t'|$ are positive. Let A^* be the maximum-coverage partial alignment of t and t' , computed in line 2 of Algorithm 1, and let

$$\begin{aligned} \text{Skipped}_{t \rightarrow t'} &= \{n \in t \mid \neg \exists n' \in t' : \mu(n, n')\}, \\ \text{Skipped}_{t' \rightarrow t} &= \{n' \in t' \mid \neg \exists n \in t : \mu(n, n')\}, \\ \text{Misordered}_{t \rightarrow t'} &= \{n_i \in t \mid (\exists n' \in t' : \mu(n_i, n')) \wedge \neg \exists j : (i, j) \in A^*\}, \\ \text{Misordered}_{t' \rightarrow t} &= \{n'_j \in t' \mid (\exists n \in t : \mu(n, n'_j)) \wedge \neg \exists i : (i, j) \in A^*\} \end{aligned}$$

be the four difference sets it computes: the skipped sets collect nodes with no μ -partner anywhere in the other trace, whereas the misordered sets collect nodes that do have a μ -partner but are left out of A^* – necessarily because pairing them with a partner would conflict with the order of the pairs already in A^* , since any pairing that did not conflict would have been included in a maximum-coverage alignment. We call a partial alignment *complete* if every node of t and every node of t' participates in it.

We can group the two penalties of Algorithm 1 into the *directional deficits*

$$\begin{aligned} \delta(t \rightarrow t') &= |\text{Skipped}_{t \rightarrow t'}| + \frac{1}{2} |\text{Misordered}_{t \rightarrow t'}|, \\ \delta(t' \rightarrow t) &= |\text{Skipped}_{t' \rightarrow t}| + \frac{1}{2} |\text{Misordered}_{t' \rightarrow t}|, \end{aligned}$$

which are exactly the quantities subtracted from $|t|$ and $|t'|$ when the algorithm forms the two directional scores. With this notation, the mean normalized difference weight of Proposition 2.2 – which counts a skipped node as 1 and a misordered node as 0.5 in each direction – is

$$\|\mathcal{D}(t, t')\|_w = \frac{1}{2} \left(\frac{\delta(t \rightarrow t')}{|t|} + \frac{\delta(t' \rightarrow t)}{|t'|} \right). \quad (1)$$

The proposition follows from three claims, proved in turn below.

Claim 1. *Every node of t satisfies exactly one of the following: (i) its index participates in A^* ; (ii) it belongs to $\text{Skipped}_{t \rightarrow t'}$; (iii) it belongs to $\text{Misordered}_{t \rightarrow t'}$. The symmetric statement holds for every node of t' .*

Proof. Let $n_i \in t$. Suppose first that i participates in A^* , so that $(i, j) \in A^*$ for some j . Condition (1) of Definition 2.1 gives $\mu(n_i, n'_j)$, so n_i has a μ -partner in t' and is therefore not skipped; and n_i is not misordered, since that would require i not to participate in A^* . Suppose instead that i does not participate in A^* . If n_i has no μ -partner in t' , then n_i is skipped and not misordered; if n_i does have a μ -partner in t' , then n_i is misordered and not skipped. The three cases are therefore exhaustive and pairwise disjoint. The argument for t' is symmetric. $\square$

In particular, $\text{Skipped}_{t \rightarrow t'}$ and $\text{Misordered}_{t \rightarrow t'}$ are disjoint subsets of t , so $0 \leq \delta(t \rightarrow t') \leq |t|$, and symmetrically $0 \leq \delta(t' \rightarrow t) \leq |t'|$. By Equation (1) this yields $\|\mathcal{D}(t, t')\|_w \in [0, 1]$, and hence, by Claim 2 below, $\text{sim}(t, t') \in [0, 1]$.

Claim 2. $\text{sim}(t, t') = 1 - \|\mathcal{D}(t, t')\|_w$.

Proof. In terms of the directional deficits, the two directional scores of Algorithm 1 read

$$\text{sim}(t \rightarrow t') = \frac{|t| - \delta(t \rightarrow t')}{|t|} = 1 - \frac{\delta(t \rightarrow t')}{|t|}, \quad \text{sim}(t' \rightarrow t) = \frac{|t'| - \delta(t' \rightarrow t)}{|t'|} = 1 - \frac{\delta(t' \rightarrow t)}{|t'|}.$$

The algorithm sets $\text{sim}(t, t')$ to the mean of the two, so

$$\text{sim}(t, t') = \frac{1}{2} \left(2 - \frac{\delta(t \rightarrow t')}{|t|} - \frac{\delta(t' \rightarrow t)}{|t'|} \right) = 1 - \frac{1}{2} \left(\frac{\delta(t \rightarrow t')}{|t|} + \frac{\delta(t' \rightarrow t)}{|t'|} \right),$$

which equals $1 - \|\mathcal{D}(t, t')\|_w$ by Equation (1). $\square$

Claim 3. $\|\mathcal{D}(t, t')\|_w = 0$ if and only if t and t' align under μ in the sense of Definition 2.1.

Proof. We argue by a chain of equivalences.

First, each directional deficit is a sum of non-negative terms, and by Equation (1) the weight $\|\mathcal{D}(t, t')\|_w$ is a sum of two non-negative terms. Hence $\|\mathcal{D}(t, t')\|_w = 0$ if and only if all four difference sets are empty.

Second, by Claim 1 every node of t and of t' is either a participant in A^* , skipped, or misordered. The four difference sets are therefore empty if and only if every node of both traces participates in A^* , that is, if and only if A^* is complete.

Third, A^* is complete if and only if t and t' align under μ . Note that a partial alignment that is complete already satisfies conditions (1) and (3) of Definition 2.1, and completeness is precisely condition (2). So if A^* is complete, it is itself an alignment

of t and t' . Conversely, suppose t and t' align via some relation A . Then A is a complete partial alignment, and since no partial alignment can cover more nodes than a complete one does, A has maximum coverage. The alignment A^* also has maximum coverage, so it covers just as many nodes as A , namely all of them; hence A^* is complete.

Combining the three equivalences proves the claim. $\square$

PROPOSITION 2.2 Let $\|\mathcal{D}(t, t')\|_w$ be the mean normalized difference weight, counting a skipped node as 1 and a misordered node as 0.5 in each direction. For all traces t, t', t'' : $\text{sim}(t, t') = 1$ exactly when $\|\mathcal{D}(t, t')\|_w = 0$, that is, when t and t' align under μ ; and $\text{sim}(t, t') < \text{sim}(t, t'')$ exactly when $\|\mathcal{D}(t, t')\|_w > \|\mathcal{D}(t, t'')\|_w$. *The proof is available online [?].*

Proof of Proposition 2.2. For the first statement, Claim 2 gives $\text{sim}(t, t') = 1$ if and only if $\|\mathcal{D}(t, t')\|_w = 0$, which by Claim 3 holds if and only if t and t' align under μ .

For the second statement, let t'' be any trace compared with t by Algorithm 1. By Claim 2,

$$\text{sim}(t, t') < \text{sim}(t, t'') \iff 1 - \|\mathcal{D}(t, t')\|_w < 1 - \|\mathcal{D}(t, t'')\|_w \iff \|\mathcal{D}(t, t')\|_w > \|\mathcal{D}(t, t'')\|_w,$$

where the second equivalence holds because subtracting both sides from 1 reverses the inequality. Hence the pairwise similarity orders trace pairs exactly inversely to their mean normalized difference weight. $\square$